Title of the Project

Design and Development of an Explainability-Driven Hallucination Detection Framework for Offline Retrieval-Augmented Generation

We are not building a chatbot.
We are building a **knowledge system** that retrieves documents, explains why they were selected, and only then generates answers.

(No cloud APIs, No ChatGPT, No black-box answers)

Problem Statement

Large Language Models (LLMs) integrated with Retrieval-Augmented Generation (RAG) frameworks have significantly improved the factual grounding of generated responses by conditioning generation on retrieved documents. However, despite retrieval support, **hallucinations i.e., generated statements that are unsupported or weakly supported by retrieved evidence remain a critical challenge**, particularly in domain-specific and offline deployments.

Most existing RAG systems primarily focus on improving retrieval quality or generation fluency, while hallucination detection is typically addressed using **external fact-checking models, entailment classifiers, or post-hoc validation techniques**. These approaches often introduce additional computational overhead, reduce interpretability, and are unsuitable for **offline or resource-constrained environments**. Moreover, hallucination detection mechanisms are commonly treated as independent modules, disconnected from the internal decision-making processes of the RAG pipeline.

At the same time, recent advances in **explainable artificial intelligence (XAI)** have enabled models to provide insight into retrieval relevance, attention patterns, similarity scores, and source attribution. However, **these explainability signals are rarely exploited as primary features for hallucination detection and control**, despite their direct relevance to assessing evidence support for generated content.

This project addresses the following research problem:

How can explainability signals generated within an offline Retrieval-Augmented Generation framework be systematically leveraged to detect, suppress, or control hallucinated outputs, while maintaining transparency, efficiency, and evidence grounding?

The problem is characterized by several challenges:

- Generated responses may include claims that are not fully supported by retrieved documents, even when retrieval quality is high.
- Existing hallucination detection methods rely on heavyweight models or external services, limiting offline applicability.
- RAG systems lack mechanisms to quantify and explain the degree of evidence support for individual generated statements.

- There is insufficient integration between retrieval explainability and generation reliability within a unified framework.

To address these challenges, this project proposes the **design and development of an explainability-driven hallucination detection framework** integrated into an offline RAG system. The framework utilizes internal explainability outputs such as document-chunk attribution, similarity heatmaps, provenance scores, and confidence measures to identify unsupported or weakly supported generated content. Based on this analysis, the system can selectively block hallucinated responses, return partial answers with explicit citations, or request query clarification.

By embedding hallucination detection within the explainability layer of the RAG pipeline, the project aims to move beyond post-hoc validation toward **intrinsic reliability assessment**, enabling the development of **transparent, trustworthy, and offline-capable knowledge systems**.

Problem Statement (in short- to use in slide)

Despite retrieval support, RAG systems can generate hallucinated responses that are weakly supported or unsupported by retrieved evidence, especially in offline settings. This project addresses the problem of **detecting and controlling hallucinations using internal explainability signals within the RAG pipeline** to ensure trustworthy, evidence-grounded generation.

Novelty:

This project introduces an **explainability-driven hallucination detection mechanism** that uses internal evidence attribution and confidence signals from a RAG pipeline. Unlike existing methods that depend on external validation models, the proposed approach enables **transparent, offline, and evidence-grounded generation**.

The novelty of this project lies in **leveraging internal explainability signals of a Retrieval-Augmented Generation (RAG) system as the primary mechanism for hallucination detection and control**. Unlike existing approaches that rely on external fact-checkers or heavyweight validation models, the proposed framework uses evidence attribution, relevance scores, and confidence measures generated within the RAG pipeline to identify unsupported outputs.

By integrating hallucination detection directly into the explainability layer of an **offline RAG system**, the project enables transparent, evidence-backed response generation with minimal computational overhead, making it suitable for resource-constrained and privacy-sensitive environments.

Objectives of the Project

- To design and implement an **offline Retrieval-Augmented Generation (RAG) framework** for domain-specific knowledge retrieval.
- To develop an **explainability layer** that provides evidence attribution and confidence scores for retrieved documents and generated responses.
- To propose an **explainability-driven hallucination detection mechanism** that identifies unsupported or weakly supported generated statements.
- To control hallucinated outputs by **suppressing responses, providing partial answers with citations, or requesting clarification**.
- To experimentally evaluate the system using **hallucination detection metrics, retrieval accuracy, and explainability quality**.